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Holmes-VAU: Towards Long-term Video Anomaly Understanding at Any Granularity

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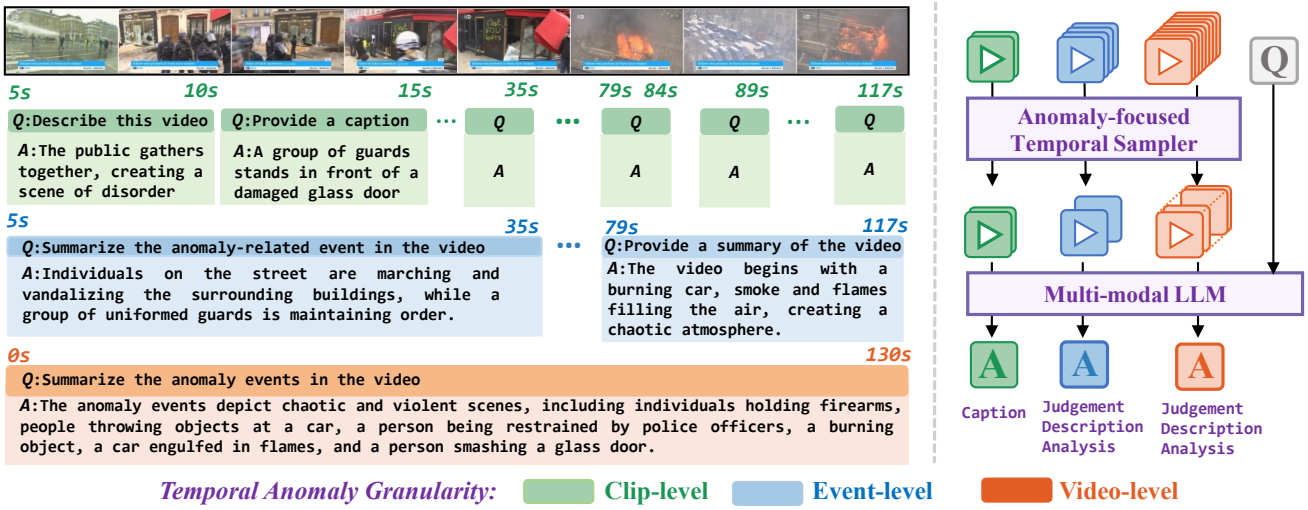

Figure 1. **Motivation**. **Left**: Existing datasets lack the hierarchical structure to capture transient and sustained anomalies across varying temporal scales. Our ***HIVAU-70k*** dataset addresses this by providing multi-granularity annotations—clip, event, and video levels—that enable detailed anomaly analysis in complex real-world scenarios. **Right**: Inspired by Sherlock Holmes's knack for zeroing in on critical details, our ***Holmes-VAU*** method integrates an Anomaly-focused Temporal Sampler with a multi-modal LLM, directing model attention to anomaly-rich segments, which enables models to decode complex, long-term video anomalies efficiently.

Abstract

How can we enable models to comprehend video anomalies occurring over varying temporal scales and contexts? Traditional Video Anomaly Understanding (VAU) methods focus on frame-level anomaly prediction, often missing the interpretability of complex and diverse real-world anomalies. Recent multimodal approaches leverage visual and textual data but lack hierarchical annotations that capture both short-term and long-term anomalies. To address this challenge, we introduce HIVAU-70k, a large-scale benchmark for hierarchical video anomaly understanding across any granularity. We develop a semi-automated annotation engine that efficiently scales high-quality annotations by combining manual video segmentation with recursive free-text annotation using large language models (LLMs). This results in over 70,000 multi-granular annotations organized at clip-level, event-level, and video-level segments. For efficient anomaly detection in long videos, we propose the Anomaly-focused Temporal Sampler (ATS). ATS integrates an anomaly scorer with a density-aware sampler to adaptively select frames based on anomaly scores, ensuring that the multimodal LLM concentrates on anomaly-rich regions, which significantly enhances both efficiency and accuracy. Extensive experiments demonstrate that our hierarchical instruction data markedly improves anomaly comprehension. The integrated ATS and visual-language model outperform traditional methods in processing long videos. Our benchmark and model are publicly available at https://github.com/pipixin321/HolmesVAU.

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1. Introduction

Video Anomaly Understanding (VAU) is crucial for applications such as video surveillance [46], violent content analysis [56], and autonomous driving [63]. Detecting deviations from normal patterns aids in hazard prevention and real-time decision-making. Traditional methods [14, 49, 74] mainly focus on frame-level predefined closed-set anomaly prediction, assigning an anomaly score to each frame. However, these approaches often fail to describe and understand complex anomalies in the real world.

To address this gap, open-world anomaly understanding [57] embraces the diversity and unpredictability of real-world anomalies. Recent work integrates multimodal approaches, combining visual data with textual descriptions [42, 58, 62, 65], while advances in multimodal visual-language models (VLMs) [6, 21, 27, 29, 68] have enabled more nuanced understanding through anomaly-related instruction tuning and text generation [9, 36, 47, 67].

Despite these advancements, **a significant gap remains in models' ability to comprehend anomalies across multiple temporal scales.** For instance, while anomalies such as explosions or fights may be captured in a single frame, more complex events like theft or arson require understanding long-term contextual patterns. Existing VAU datasets [46, 56] typically provide annotations at a single level of granularity, limiting models to understanding either immediate perceptual anomalies or those requiring extended contextual reasoning. The lack of datasets with hierarchical annotations—encompassing both short-term and long-term anomalies—hinders models' capacity to reason about anomalies with diverse temporal characteristics. Moreover, constructing datasets that encapsulate this hierarchical complexity poses significant challenges in scalability and annotation quality.

To address these issues, we develop a semi-automated annotation engine that efficiently scales high-quality annotation by combining manual video segmentation with recursive free-text annotation using large language models (LLMs). The process involves three key stages: **1)** *hierarchical video decoupling*, where we manually identify anomaly events and segment them into shorter clips; **2)** *hierarchical free-text annotation*, where captions for each clip are generated through human effort or video captioning models, then summarized at the event and video levels via LLMs; and **3)** *hierarchical instruction construction*, where the textual data is transformed into question-answer instruction prompts by combining captions and summaries with designed prompts, creating a dataset with rich annotations for training and evaluating models.

Utilizing the annotation engine, we introduce ***HIVAU-70k***, a large-scale video anomaly understanding benchmark with hierarchical instructions. Our dataset comprises over 70,000 multi-granular instruction data organized across clip-level, event-level, and video-level segments as shown in Fig. 1. This hierarchical structure empowers models to detect immediate anomalies, *e.g.*, sudden explosions or fighting, as well as complex events that require an understanding of long-term context, like theft or arson. By annotating at multiple temporal levels, HIVAU-70k provides diverse anomalies in open-world scenarios.

Towards long-term VAU, efficiency remains a critical challenge. Previous methods [9, 47, 67] often rely on *uniform frame sampling*, which can either miss crucial anomaly frames or incur large computational costs [25, 47, 67]. To address this, we propose the ***Holmes-VAU*** method, which combines the proposed Anomaly-focused Temporal Sampler (ATS) with the multimodal visual-language model for efficient long-term video anomaly understanding (See Fig. 1). The ATS combines a *anomaly scorer* with a *density-aware sampler* that adaptively selects frames by their anomaly scores. This integration ensures that the visual-language model concentrates on anomaly-rich regions, enhancing both efficiency and accuracy.

Our contributions are threefold: **1)** We introduce ***HIVAU-70k***, a large-scale, multi-granular benchmark for hierarchical video anomaly understanding. **2)** We propose the ***Holmes-VAU*** method, which combines the proposed Anomaly-focused Temporal Sampler (ATS) to boost the efficiency and accuracy of inference on long-term videos. **3)** We conduct extensive experiments demonstrating the effectiveness of hierarchical instruction data in enhancing anomaly comprehension and validate the performance gains provided by the integrated ATS and visual-language model in processing long videos.

2. Related Works

Video Anomaly Detection. This task aims to temporally detect abnormal frames in a long untrimmed video [1, 14, 23, 34, 40, 52]. Early attempts are based on handcrafted features [1, 19, 23, 34, 40, 72]. Recently, deep learning approaches [14, 37, 61] have become dominant, broadly classified into unsupervised, weakly-supervised, and fully-supervised methods. Unsupervised methods train only on normal videos to learn normal patterns and are often designed as reconstruction-based [13, 14, 59, 61], prediction-based [31], or a combination [33]. Some methods [48, 50, 66] also explore a fully unsupervised setting, including both normal and abnormal videos in training set without real labels. Weakly-supervised methods [11, 22, 37, 46, 49, 55, 56, 70, 73, 74] use both normal and abnormal videos with video-level annotations. Fully-supervised methods [20, 30] are less studied due to the high cost of precise frame-level annotations.

Multi-modal Video Anomaly Understanding. Large-scale visual-language pre-trained models such as CLIP [44]

serve as a bridge between visual and textual modalities. Some recent works [17, 42, 58, 62] in the realm of video anomaly detection have leveraged textual information as prompts to enhance the model's anomaly representation. Based on this, [57] firstly proposed the open vocabulary VAD task, [65] introduced a multimodal video anomaly dataset composed of dense clip captions. Furthermore, [67] extracted captions from video frames and designed prompts for LLMs to provide anomaly scores, [9] and [47] construct diverse and interactive instruction data at the video level. However, these datasets consider only a single temporal level of anomaly understanding data construction, *i.e.* clip-level [65] or video-level [9, 47]. Unlike these methods, we focus on building large-scale *hierarchical* video anomaly understanding data for multimodal instruction tuning.

Hierarchical Video Understanding. Video understanding is a challenging task due to its temporal-scale diversity. To better comprehend videos, many previous works have focused on both datasets and models in hierarchical video understanding. For example, [3, 24, 32, 35, 45, 54] proposed fine-grained action recognition and localization datasets, [16] provided free-form hierarchical captions for hour-long videos at multiple temporal scales, and [8, 18, 38, 43, 60, 69] trained models on hierarchical levels to obtain better video feature representation. Recently, to assess the capability of video vision-language models (VLMs) in handling challenges in real-world scenarios, several video benchmarks [10, 12] have been built, incorporating data at multiple temporal scales for evaluation. Unlike these works, we dive into the field of Video Anomaly Understanding, thus filling the gap in multi-scale annotations in this area.

3. HIVAU-70k Benchmark

We first define the video anomaly understanding task in Sec.3.1. Then, the construction process of the HIVAU-70k benchmark will be elaborated in Sec.3.2. Finally, we will present HIVAU-70k's statistical information in Sec. 3.3.

3.1. Task Description

This work focuses on video anomaly understanding, involving temporal anomaly detection and anomaly explainability. Temporal anomaly detection aims to predict an anomaly score for each frame in a video $\mathcal{V}$, represented as $S \in \mathbb{R}^T$, where T is the total number of frames. Building on this, we explore the model's ability to generate explanatory text outputs related to anomalies based on video input and user queries. Specifically, given a video $\mathcal{V}$ and a text query $\mathcal{Q}$, the model generates an anomaly-related response $\mathcal{A}$. We consider two key abilities: **1**) **visual perception**, which involves recognizing main entities in the video, and **2**) **anomaly reasoning**, which encompasses the model's judgment and analysis of the anomaly content.

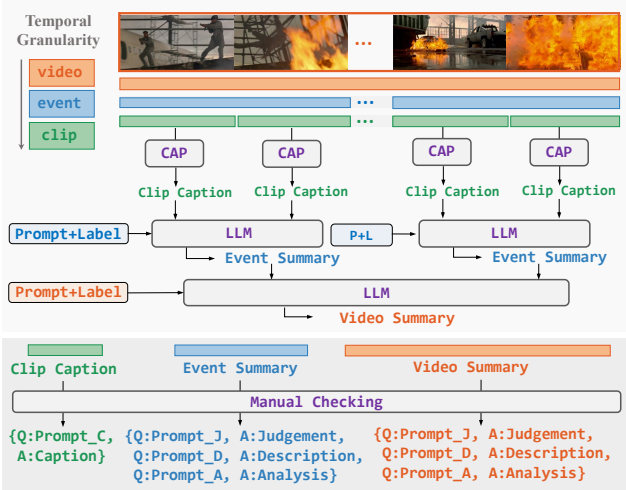

Figure 2. **Data Engine**. We present a structured workflow for generating hierarchical annotations across video, event, and clip levels. Clips are first captioned, then processed through a large language model (LLM) with prompts for event summarization. The outputs include clip captions, event summaries, and video summaries, followed by manual checking and refinement. This multi-step approach enriches the dataset with detailed judgments, descriptions, and analyses of anomalies, enabling robust contextual understanding at varying granularities.

3.2. LLM-Empowered Data Engine

As shown in Fig.2, we develop a semi-automated annotation engine that efficiently scales high-quality annotations, which consists of three main steps: 1) hierarchical video decoupling, 2) hierarchical free-text annotation, and 3) hierarchical instruction construction.

Hierarchical Video Decoupling. Our video sources include the training set of the UCF-Crime [46] dataset and the XD-Violence [56] dataset, which contains videos of varying durations and diverse real-world anomalies. For abnormal videos, we first manually obtain the temporal boundaries of each anomaly event in the video. Non-continuous anomalous states are considered separate events. Then, we divide each event into clips of random lengths. For normal videos, we apply random sampling to obtain corresponding segments of varying granularities. Ultimately, we obtained 5,443 videos, 11,076 events, and 55,806 clips. This process took 5 annotators approximately 20 hours to complete. More details can be found in the appendix.

Hierarchical Free-text Annotation. To fully extract semantic information from the clip-level videos, we utilize a powerful off-the-shell video perception model LLaVA-Next-Video [28] to generate detailed captions for each clip. We also include the UCA dataset [64], which provides manually annotated captions for video clips in the UCF-Crime [46] dataset. Then, we use an LLM [2] to consoli-

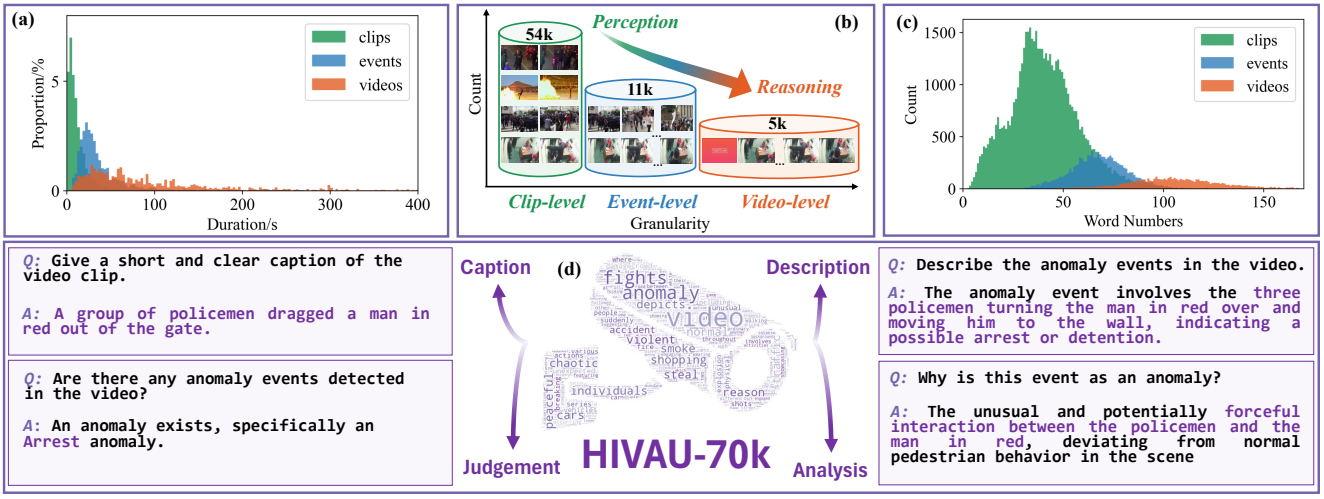

Figure 3. **HIVAU-70k dataset**. (**a**) Duration distributions for clips, events, and full videos, showing dominance of short clips. (**b**) Hierarchical data organization from clip-level to video-level, enabling perception-to-reasoning insights. (**c**) Word count variations across annotation levels, with more detailed descriptions at the video level. (**d**) Sample annotations capturing captioning, judgment, description, and anomaly analysis, highlighting nuanced understanding of anomaly events in complex scenes.

date all clip captions within an event, generating an event-level video summary. Specifically, we design prompts[1] to guide the LLM to produce three parts of content for each anomalous event summary: 1) *Judgment*: A determination of whether an anomaly exists and its specific category, 2) *Description*: A detailed description of the anomalous or normal event, 3) *Analysis*: The reasoning behind the anomaly judgment, including causal analysis. To guide the LLM in generating reliable responses, we also inject the event's category label (*e.g.*, "Shooting", "Explosion") into the LLM prompt. We consolidate all event-level summaries to obtain the video-level summary. This results in free-text annotations across multiple temporal scales, including short-term visual perception (clip-level) and long-term anomaly reasoning (event-level, video-level).

Hierarchical Instruction Data Construction. The ability of VLMs to follow user instructions and generate responses is achieved through instruction-tuning [21, 25, 27, 29]. The training data format typically consists of the following:

'{$\mathcal{Q}$:*user instruction*, $\mathcal{A}$:*model response.*}'

To build instruction-tuning data for VLMs in the domain of Video Anomaly Understanding, we matched free-text annotations with pre-designed anomaly-related user instructions. Specifically, for clip-level segments, we only construct instructions related to captions, as it is challenging to obtain anomaly-related analysis for short videos. For event-level and video-level segments, we construct instruction data from the perspectives of Judgment, Description, and Analysis. Typical examples are shown in the Fig. 3 (d).

Manual Checking for Data Quality Control. To ensure dataset quality, we implemented several human inspection and curation strategies. First, we labeled the temporal boundaries of abnormal event segments and reviewed them at the second level. Next, anomaly labels were incorporated during summary generation using LLMs, directing focus on relevant entities. Finally, manual reviews were performed to correct low-quality instruction data.

[1]For detailed prompts, please refer to the appendix.

3.3. Data Statistic of HIVAU-70k

Utilizing the proposed annotation engine, we introduce HIVAU-70k, as shown in Fig. 3, a large-scale benchmark designed for hierarchical instruction-based video anomaly understanding. As shown in Fig. 3(b), HIVAU-70k contains over 70,000 multi-granular annotations organized at clip-level, event-level, and video-level segments, achieving a progression from perception to reasoning. As shown in Fig. 3(a) and (c), the durations of segments and the word numbers of text annotations at different granularities exhibit significant distributional differences. As shown in Fig. 3(d), HIVAU-70k's instruction data covers *Caption*, *Judgment*, *Description*, and *Analysis* for real-world anomalies, which guide the model to develop both short-term and long-term video anomaly understanding capabilities.

4. Method: Holmes-VAU

Long-term video anomaly understanding with LLMs/VLMs has traditionally been hindered by frame redundancy, complicating accurate anomaly detection. Previous VAU approaches struggle with focus: methods like dense window sampling [67] add redundancy, and uniform frame sampling [9, 47] often misses key anomalies, limiting application to short videos. We introduce the Anomaly-focused Temporal Sampler (ATS) to address this,

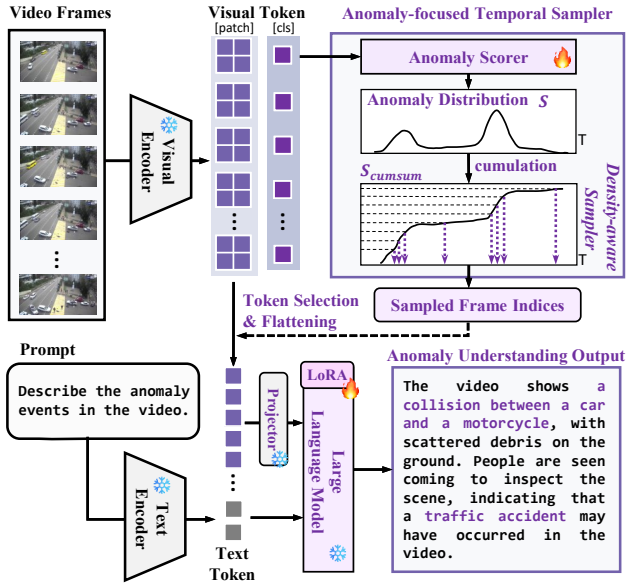

Figure 4. **Holmes-VAU: a multi-modal-LLM-based video anomaly detection framework with adaptive anomaly focus.**

integrating it into the VLM, and fine-tuning it via instruction on HIVAU-70k to form our Holmes-VAU model.

4.1. Pipeline Overview

The overall pipeline of our Holmes-VAU model is shown in Fig. 4. Video frames are processed by a visual encoder, creating visual tokens. These tokens are analyzed by an Anomaly-focused Temporal Sampler using an anomaly scorer and cumulative sum (S_{cumsum}) to select keyframes. A text encoder processes a prompt (e.g., 'Describe the abnormal events in the video'). Visual and textual representations are combined in a pre-trained language model, fine-tuned with LoRA, to generate a description of detected anomalies, such as a 'collision between a car and a motorcycle' and 'traffic accident' indicators.

4.2. Model Architecture

Visual and Text embedding. We utilize the frozen visual encoder in InternVL2 [6], which inherits the ViT structure from CLIP [44], we refer to it as ϕ_v. Following previous VAU works [55, 58, 67, 74], we sample dense video frames at an interval of 16 frames from the input video. Each video frame is then processed through the visual encoder to obtain the corresponding visual tokens. Given the input video frame sequence $\mathcal{V} \in \mathbb{R}^{T\times H\times W\times C}$, the output features of i-th frame can be denotes as:

$$V_i = \{v_i^{cls}, v_i^1, v_i^2, ..., v_i^{N_p}\} = \phi_v(\mathcal{V}_i) \tag{1}$$

where v_i^{cls} indicates the class token, v_i^j ($j \in \{1, 2, ..., N_p\}$) denotes the patch tokens, and N_p reperesents the number of patches. The text encoder ϕ_t is also initialized from [6], which includes a tokenizer and an embedding layer. The prompt text $\mathcal{Q}$ is converted into text tokens through the text encoder: $X_q = \phi_t(\mathcal{Q})$.

Anomaly-focused Temporal Sampler (ATS). ATS consists of two components: the *anomaly scorer* and the *density-aware sampler*. The *anomaly scorer* ϕ_s is a feature-based VAD network which estimates the anomaly score for each frame. We follow the network architecture in [74] due to its simplicity and good performance. Given the class token of the video frames $\{v_1^{cls}, v_2^{cls}, ..., v_T^{cls}\}$, the anomaly scores can be obtained: $s_i = \phi_s(v_i^{cls})$, where s_i denotes the anomaly score of the i-th frame.

Anomalous frames typically contain more information and exhibit greater variation than normal frames [49]. This observation motivates us to sample more frames in regions with higher anomaly scores while reducing the sampling in areas with lower anomaly scores. As shown in Fig. 4, to achieve non-uniform sampling, we propose *density-aware sampler* to selectively choose N frames from a total of T input frames. Specifically, we treat the anomaly scores $S \in \mathbb{R}^T$ as a probability mass function and first accumulate them along the temporal dimension to obtain the cumulative distribution function, denoted as S_{cumsum}:

$$S_{cumsum}(t) = \sum_{i=1}^{t}(s_i + \tau) \tag{2}$$

We uniformly sample N points along the cumulative axis, then map these points to the cumulative distribution S_{cumsum}, the corresponding N timestamps on the time axis are mapped to the closest frame index and finally form the sampled frame indices, denoted as $\mathcal{G}$. τ is used to control the uniformity of the sampling.

Projector and LLM. We select the tokens corresponding to the sampled frame, *i.e.*, $\mathcal{G}$, as the visual embedding. A projector ϕ_p is then used to map the visual embedding to the language feature space. Finally, we concatenate these embeddings with the text embeddings, input them into the pre-trained large language model, and compute the probability of the target answers X_a. To obtain an initial visual-language alignment space, we initialize the projector and LLM parameters from [6], with the parameters kept frozen. The above process can be expressed as follows:

$$X_{ins} = \text{cat}[\phi_p(\text{cat}[V_i]), X_q] \qquad i \in \mathcal{G} \tag{3}$$

$$p(X_a|X_{ins}) = \prod_{i=1}^{L} p_\theta|(x_i|X_{ins,<i}, X_{a,<i}) \tag{4}$$

where cat[·] represents the concatenation operation, θ is the trainable parameters, $X_{ins,<i}, X_{a,<i}$ are the instruction and answer tokens in all turns before the current prediction token x_i, L is the length of sequence, respectively.

4.3. Training and Testing

Training. We train the model following two steps. In the first step, we use the video data and annotated frame-level label ($\hat{y} \in \mathbb{R}^T$) from HIVAU-70k to train the *anomaly scorer*, which provides more accurate anomaly supervision compared to previous unsupervised and weakly-supervised methods [14, 46, 55, 74].

$$\mathcal{L}_{AS} = -\sum_{i=1}^{T}(s_i log(\hat{y}_i) + (1 - s_i) log(1 - \hat{y}_i)) \quad (5)$$

In the second step, we keep the anomaly scorer fixed, and use all the instruction data from HIVAU-70k to train the model. To achieve more efficient fine-tuning without disrupting the original capabilities of the LLM, we employ LoRA [15] for fine-tuning, optimizing the cross entropy loss between the predicted and the ground truth tokens.

Testing. During testing, the user inputs a video and text prompts; the model will generate the corresponding text response following the user's instruction.

5. Experiments

5.1. Experiment Setup

Dataset. Our HIVAU-70k is built upon two large-scale real-world datasets, *i.e.*, UCF-Crime [46] and XD-Violence [56], they provide a diverse range of videos with anomalous events. UCF-Crime [46] comprises 1,900 untrimmed videos totaling 128 hours from outdoor and indoor surveillance cameras. It encompasses 13 classes of real-world anomalies, including *Abuse*, *Explosion*, *Fighting*, and *Shooting*. XD-Violence [56] is the largest VAD benchmark, comprising 4,754 videos totaling 217 hours sourced from surveillance, movies, car cameras, and games. It encompasses 6 anomaly classes: *Abuse*, *Car Accidents*, *Explosions*, *Fighting*, *Riots*, and *Shooting*.

Metric. We assess the anomaly understanding ability from two aspects: ***anomaly detection*** and ***reasoning***. **1**) For ***anomaly detection***, we use the anomaly scores output by the Anomaly Scorer as the prediction and perform the evaluation. Following [14, 46, 67, 74], we use AUC and AP to quantify detection performance, which is evaluated only on the video level. **2**) For ***anomaly reasoning***, we annotate instruction data from the UCF-Crime and XD-Violence test sets, which have been carefully reviewed and filtered by annotators. We finally collected 3,300 test samples at multiple granularities. The test set contains 2200/732/398 samples at clip/event/video levels. We calculate metrics including BLEU [41], CIDEr [51], METEOR [4] and ROUGE [26] to measure the quality of the reasoning text output by the model, comparing with the annotated ground truth text.

Implementation Details. For the proposed Holmes-VAU method, we initialize the Multimodal LLM with InternVL2-2B [6]. To optimize the Anomaly-focused Temporal Sampler, we adopt the Adam optimizer with a learning rate of 1e-4. Note that when evaluating detection performance on XD-Violence and UCF-Crime, only videos in the corresponding training sets are used to train our model for fair comparisons. For instruction tuning, we train with a batch size of 512 for 1 epoch, using the AdamW optimizer with cosine learning rate decay and a warm-up period. The LoRA [15] parameters are set as: r=64, α=128, and learning rate=4e-5. During testing, τ in Eq. 2 is set to 0.1. Experiments are conducted on 2 NVIDIA A100 GPUs.

Table 1. **Comparison of detection performance with state-of-the-art Video Anomaly Detection approaches**. We include the results of explainable and non-explainable methods. "$*$" represents the result reported in [67].

Methods	Backbone	XD-Violence	UCF-Crime
		AP/%	AUC/%
Non-explainable VAD			
Conv-AE [14] (CVPR'16)	-	27.25	50.60
GODS [52] (ICCV'19)	I3D	N/A	70.46
GCL [66] (CVPR'22)	ResNext	N/A	71.04
DYANNET [48] (WACV'23)	I3D	N/A	84.50
MIST [11] (CVPR'21)	I3D	N/A	82.30
Wu *et al.* [56] (ECCV'20)	I3D	78.64	82.44
RTFM [49] (ICCV'21)	I3D	77.81	84.30
MSL [22] (AAAI'22)	I3D	78.28	85.30
S3R [55] (ECCV'22)	I3D	80.26	85.99
MGFN [5] (AAAI'23)	I3D	79.19	86.98
UR-DMU [74] (AAAI'23)	I3D	81.66	86.97
CLIP-TSA [17] (ICIP'23)	ViT	82.19	87.58
VadCLIP [58] (AAAI'24)	ViT	84.51	88.02
Yang *et al.* [62] (CVPR'24)	ViT	83.68	87.79
Wu *et al.* [57] (CVPR'24)	ViT	66.53	86.40
Explainable Multi-modal VAD			
Zero-Shot CLIP [44]*	ViT	17.83	53.16
LLAVA-1.5 [27]*	ViT	50.26	72.84
LAVAD [67] (CVPR'24)	ViT	62.01	80.28
***Holmes*-VAU (Ours)**	ViT	**87.68**	**88.96**

5.2. Main Results

Anomaly Detection Results. We compare our method with state-of-the-art methods, including semi-supervised methods [14, 52], unsupervised methods [48, 66], weakly-supervised methods [17, 22, 49, 55, 58, 74] and recently training-free method [67]. We have indicated their backbones and performance on the UCF-Crime and XD-Violence datasets, as shown in Table 1. Our method has an AP of 87.68% on XD-Violence and an AUC of 88.96% on UCF-Crime, significantly outperforming the prior state-of-the-art methods, which demonstrates that our method can generate less biased anomaly scores. It is worth noting that while achieving precise localization of anomalies, Holmes-VAU is also capable of providing explanations and analysis for the detected anomalies by the model, a feature unavailable in existing non-explainable VAD methods. Although LAVAD [67] has explainability, the training-free large lan-

Table 2. **Comparison of reasoning performance with state-of-the-art Multimodal Large Language Models (MLLMs).** 'BLEU' refers to the cumulative values from BLEU-1 to BLEU-4. We evaluate the quality of the generated text at different granularities, including clip-level (C), event-level (E), and video-level (V).

Method	Params	BLEU [41](↑)			CIDEr [51](↑)			METEOR [4](↑)			ROUGE [26](↑)		
		C	E	V	C	E	V	C	E	V	C	E	V
Video-ChatGPT [39]	7B	0.152	0.068	0.066	0.033	0.011	0.013	0.102	0.069	0.044	0.153	0.048	0.079
Video-LLaMA [68]	7B	0.151	0.079	0.104	0.024	0.014	0.017	0.112	0.076	0.057	0.156	0.067	0.090
Video-LLaVA [25]	7B	0.164	0.046	0.055	0.032	0.009	0.013	0.097	0.022	0.014	0.132	0.023	0.045
LLaVA-Next-Video [71]	7B	0.435	0.091	0.120	0.102	0.015	0.031	0.117	0.085	0.096	0.198	0.080	0.106
QwenVL2 [53]	7B	0.312	0.082	0.155	0.044	0.020	0.044	0.133	0.092	0.112	0.163	0.081	0.137
InternVL2 [6]	8B	0.331	0.101	0.145	0.052	0.022	0.035	0.141	0.095	0.101	0.182	0.102	0.122
Holmes-VAU (Ours)	2B	**0.913**	**0.804**	**0.566**	**0.467**	**1.519**	**1.437**	**0.190**	**0.165**	**0.121**	**0.329**	**0.370**	**0.355**

Table 3. **Ablation of hierarchical instruction data.** During the instruction tuning phase, we combined training data of different granularities, including clip (C), event (E), and video (V) levels.

Training Data			BLEU(↑)			CIDEr(↑)		
C	E	V	C	E	V	C	E	V
✓			**0.984**	0.261	0.351	0.459	0.120	0.106
	✓		0.508	0.576	0.292	0.097	1.183	0.872
		✓	0.280	0.222	0.279	0.039	0.708	0.884
✓	✓		0.889	0.741	0.349	0.470	1.285	0.889
✓		✓	0.906	0.341	0.522	**0.472**	0.962	1.093
	✓	✓	0.394	0.797	0.505	0.081	1.472	1.074
✓	✓	✓	0.913	**0.804**	**0.566**	0.467	**1.519**	**1.437**

guage model lacks an understanding of anomaly knowledge due to the limitation of insufficient supervised data.

Anomaly Reasoning Results. We compare the anomaly-related text quality generated by Holmes-VAU with that produced by state-of-the-art general Multimodal Large Language Models (MLLMs), and presented the results at different temporal granularities, including clip-level, event-level, and video-level, as shown in Table. 2. Earlier MLLMs such as Video-ChatGPT [21] and Video-LLaMA [68], struggled with basic visual perception and instruction-following capabilities. Recent MLLMs [6, 7, 71] trained on larger and higher-quality video instruction data have made significant progress in general video understanding, with noticeable improvements at the clip-level perception task. However, due to the absence of learning from complex, real-world anomaly data, their reasoning abilities at the event-level and video-level are still lacking. Our Holmes-VAU, however, shows significant improvements in video understanding across all temporal granularities compared to existing general MLLMs, highlighting the importance of injecting anomaly-related knowledge through instruction tuning on high-quality Video Anomaly Understanding benchmarks.

5.3. Analytic Results

Influence of Hierarchical Instruction. To explore the impact of different granularity video training data on the model's anomaly reasoning ability, we designed various training data combinations during the instruction tuning phase and evaluated the model's performance, as shown in Table 3. The inclusion of clip-level data primarily enhanced the model's basic visual perception abilities regarding actions and scenes within the video. Adding event-level data improved the model's ability to judge and understand complete anomaly events. Furthermore, the involvement of video-level data further enhanced the model's ability to analyze and summarize anomaly-related information across longer-span videos. The hierarchical instruction data structure facilitated a comprehensive and complementary improvement in the model's anomaly-related perception-to-reasoning capabilities.

Table 4. **Ablation study of sampling methods and the number of sampled frames.** We compare the proposed Anomaly-focused Temporal Sampler (ATS) with other sampling methods under different frame sampling numbers, including *Uniform* and *Top-K*. Latency is the time delay in generating the first token.

Frames (N)	Sampler	Latency (ms)	Video-level	
			BLEU(↑)	CIDEr(↑)
8	*Top-K*	244	0.462	1.229
	Uniform		0.491	1.276
	ATS (Ours)		**0.514**	**1.324**
16	*Top-K*	566	0.476	1.302
	Uniform		0.511	1.345
	ATS (Ours)		**0.566**	**1.437**
32	*Top-K*	1402	0.481	1.332
	Uniform		0.558	1.357
	ATS (Ours)		**0.576**	**1.460**

Influence of different sampling methods and the number of sampled frames. Our ATS (Anomaly-focused Temporal Sampler) is designed to adaptively sample frames input to the LLM based on the frame-level anomaly scores. To validate its advantages, we compared ATS with other sampling methods at various sample frame counts, including *Uniform* and *Top-K* sampling. In *Uniform* sampling, N frames are uniformly sampled from all frames, while Top-K sampling selects the frames with the top N highest anomaly scores. As shown in Table 4, ATS consistently outperforms other sampling methods, regardless of the sample count. We believe that *Uniform* sampling tends to overlook key anomaly frames, though this issue lessens as more frames are sampled. Besides, *Top-K* sampling tends to overly focus on local anomaly frames, missing contextual frame informa-

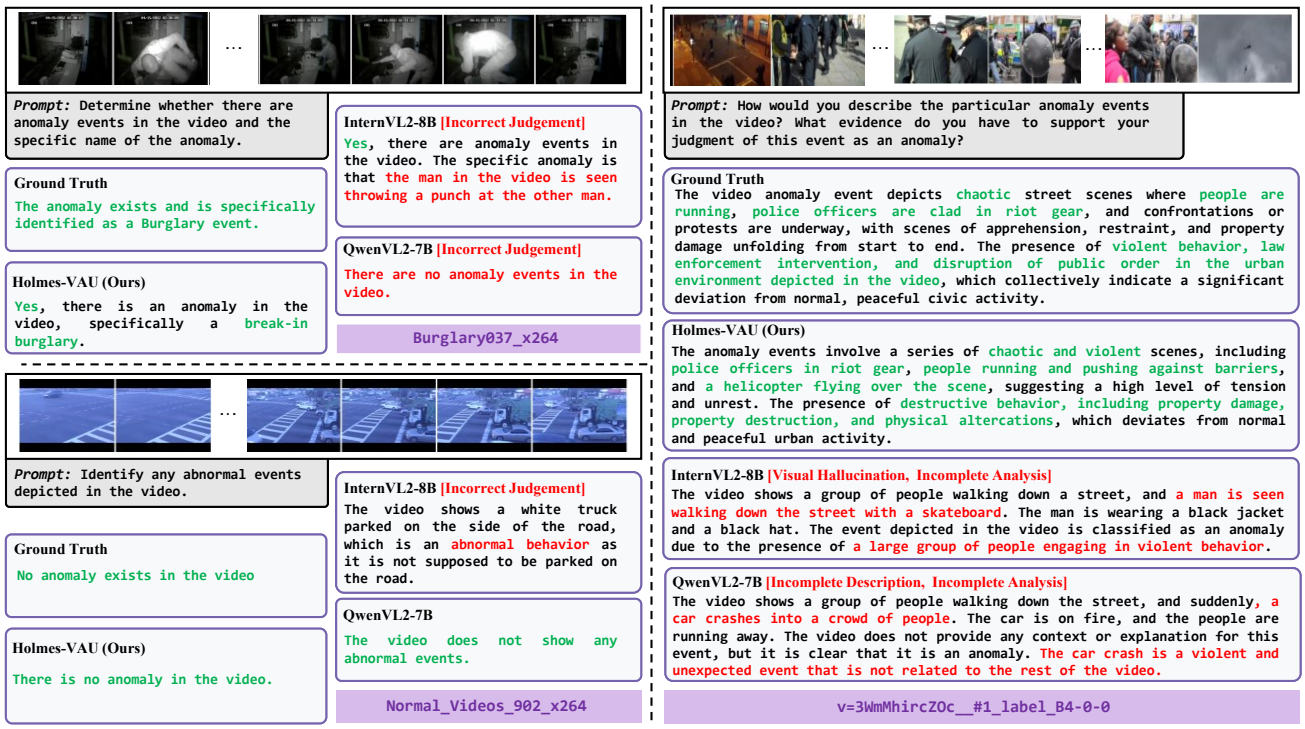

Figure 5. **Qualitative comparison of anomaly understanding explanation.** Compared with state-of-the-art general MLLMs, *i.e.*, InternVL2 [6] and QwenVL2 [53], our proposed Holmes-VAU demonstrates more accurate anomaly judgment, along with more detailed and comprehensive anomaly-related descriptions and analysis. Correct and wrong explanations are highlighted in green and red, respectively.

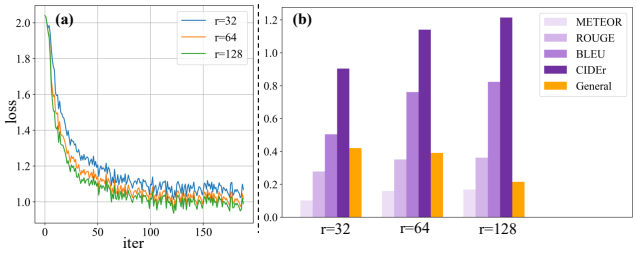

Figure 6. **Ablation study of trainable parameters.** **(a)** Loss curve during instruction-tuning. **(b)** We tuned the LoRA [15] parameter r to control trainable parameters, evaluating its impact on VAU capability, and *General* performance on Video-MME [12].

tion. Our proposed ATS mitigates both issues. To balance inference efficiency and performance, we set N=16 as the default sample frame number.

Instruction Tuning Parameters. We conducted an ablation study on the parameter r in LoRA [15] to explore how the trainable parameters affects both the model's VAU performance and its general capability. We use Video-MME [12] to evaluate the model's general capability. The results are shown in Fig. 6, as r increases, the model gradually adapts to the VAU task. However, when r becomes too large, the model's general capability decreases. To retain the original general video understanding capability of the MLLM, we set r=64 as the default value.

5.4. Qualitative Comparision

We provide qualitative comparisons between Holmes-VAU and existing MLLMs in Fig. 5. The results demonstrate that Holmes-VAU can accurately identify anomalies in videos and provide accurate and complete explanations, highlight the effectiveness and advantage of Holmes-VAU in perceiving video events and analyzing anomalies.

6. Conclusion

In conclusion, this work pushes the boundaries of video anomaly understanding by introducing hierarchical anomaly detection across diverse temporal scales, from momentary clips to extended events. The HIVAU-70k benchmark, with over 70,000 multi-level annotations, addresses a critical gap in the field, enabling comprehensive anomaly analysis in real-world scenarios. Our Anomaly-focused Temporal Sampler (ATS) strategically enhances focus on anomaly-dense segments, optimizing both efficiency and accuracy in long-term anomaly detection. Extensive experiments demonstrate that our hierarchical dataset and ATS-enhanced VLM achieve significant performance gains over conventional methods, proving robust for open-world anomaly understanding. This work sets a new standard for multi-granular anomaly comprehension, paving the way for more fine-grained video anomaly understanding.

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